Percentage:

1. **Concept of Percentage:**

   By a certain **percent**, we mean that many hundredths.

   Thus, x percent means x hundredths, written as x%.

   To express x% as a fraction: We have, $x\% = \frac{x}{100}$.

   Thus, $20\% = \frac{20}{100} = \frac{1}{5}$.

   To express $\frac{a}{B}$ as a percent: We have, $\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$.

   Thus, $\frac{1}{4} = \left(\frac{1}{4} \times 100\right)\% = 25\%$.

2. **Percentage Increase/Decrease:**

   If the price of a commodity increases by R%, then the reduction in consumption so as not to increase the expenditure is:

$$\left[\frac{R}{(100 + R)} \times 100\right]\%$$

   If the price of a commodity decreases by R%, then the increase in consumption so as not to decrease the expenditure is:

$$\left[\frac{R}{(100 - R)} \times 100\right]\%$$

3. **Results on Population:**

Let the population of a town be P now and suppose it increases at the rate of R% per annum, then:

1. Population after *n* years = $P\left(1 + \frac{R}{100}\right)^n$
2. Population *n* years ago = $\frac{P}{\left(1 + \frac{R}{100}\right)^n}$

4. **Results on Depreciation:**

Let the present value of a machine be P. Suppose it depreciates at the rate of R% per annum. Then:

1. Value of the machine after *n* years = $P\left(1 - \frac{R}{100}\right)^n$
2. Value of the machine *n* years ago = $\frac{P}{\left(1 - \frac{R}{100}\right)^n}$
3. If A is R% more than B, then B is less than A by $\left[\frac{R}{(100 + R)} \times 100\right]\%$.
4. If A is R% less than B, then B is more than A by $\left[\frac{R}{} \times 100\right]\%$

Ex. 3. Express each of the following as rate percent :

(i) $\frac{23}{36}$ (ii) $6\frac{3}{4}$ (iii) 0.004

Sol. (i) $\frac{23}{36} = \left(\frac{23}{36} \times 100\right)\% = \left(\frac{575}{9}\right)\% = 63\frac{8}{9}\%.$

(ii) $0.004 = \frac{4}{1000} = \left(\frac{4}{1000} \times 100\right)\% = 0.4\%.$

(iii) $6\frac{3}{4} = \frac{27}{4} = \left(\frac{27}{4} \times 100\right)\% = 675\%.$

Ex. 4. Evaluate :

(i) 28% of 450 + 45% of 280 (Bank P.O. 2003)

(ii) $16\frac{2}{3}\%$ of 600 gm − $33\frac{1}{3}\%$ of 180 gm (R.R.B. 1998)

Sol. (i) $28\% \text{ of } 450 + 45\% \text{ of } 280 = \left(\frac{28}{100} \times 450 + \frac{45}{100} \times 280\right) = (126 + 126) = 252.$

(ii) $16\frac{2}{3}\% \text{ of } 600 \text{ gm} - 33\frac{1}{3}\% \text{ of } 180 \text{ gm}$

$= \left[\left(\frac{50}{3} \times \frac{1}{100} \times 600\right) - \left(\frac{100}{3} \times \frac{1}{100} \times 180\right)\right] \text{gm} = (100 - 60) \text{ gm} = 40 \text{ gm}.$

(S.S.C. 2000)

Ex. 7. *Which is greatest in $16\frac{2}{3}\%$, $\frac{2}{15}$ and 0.17 ?*

Sol. $16\frac{2}{3}\% = \left(\frac{50}{3} \times \frac{1}{100}\right) = \frac{1}{6} = 0.166$, $\frac{2}{15} = 0.133$. Clearly, 0.17 is the greatest.

Ex. 11. *Difference of two numbers is 1660. If 7.5% of one number is 12.5% of the other number, find the two numbers.*

Sol. Let the numbers be x and y. Then, 7.5% of x = 12.5% of y $\Leftrightarrow$ $x = \frac{125}{75}y = \frac{5}{3}y$.

Now, $x - y = 1660 \Rightarrow \frac{5}{3}y - y = 1660 \Rightarrow \frac{2}{3}y = 1660 \Rightarrow y = \left(\frac{1660 \times 3}{2}\right) = 2490$.

$\therefore$ One number = 2490, Second number = $\frac{5}{3}y = 4150$.

A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?

A. 45%

B. $45\frac{5}{11}\%$

C. $54\frac{6}{11}\%$

D. 55%

Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are:

A. 39, 30

B. 41, 32

C. 42, 33

D. 43, 34

What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?

A. 1

B. 14

C. 20

D. 21

A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:

A. 588 apples

B. 600 apples

C. 672 apples

D. 700 apples

A fruit seller had some oranges. He sells 30% oranges and still has 140 mangoes. Originally, he had:

A. 288 oranges

B. 300 oranges

C. 672 oranges

D. 200 oranges

Ex. 8. *If the sales tax be reduced from $3\frac{1}{2}\%$ to $3\frac{1}{3}\%$, then what difference does it make to a person who purchases an article with marked price of Rs. 8400 ?*

(S.S.C. 2002)

Sol. Required difference $= \left(3\frac{1}{2}\% \text{ of Rs. } 8400\right) - \left(3\frac{1}{3}\% \text{ of Rs. } 8400\right)$

$$= \left(\frac{7}{2} - \frac{10}{3}\right)\% \text{ of Rs. } 8400 = \frac{1}{6}\% \text{ of Rs. } 8400$$

$$= \text{Rs. } \left(\frac{1}{6} \times \frac{1}{100} \times 8400\right) = \text{Rs. } 14.$$

Ex. 10. *Sixty-five percent of a number is 21 less than four-fifth of that number. What is the number ?*

Sol. Let the number be x.

Then, $\frac{4}{5}x - (65\% \text{ of } x) = 21 \Leftrightarrow \frac{4}{5}x - \frac{65}{100}x = 21 \Leftrightarrow 15x = 2100 \Leftrightarrow x = 140.$

In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:

2700

B. 2900

C. 3000

D. 3100

Number of Valid votes=80% of 7500

=6000

Valid votes polled by other candidate= 45% of 6000

=(45/100) X 6000

=2700

Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?

A. 57%

B. 60%

C. 65%

D. 90%

Total number of votes polled=(1136+7636+11628)= 20400

Required Percentage= (11628/20400)X100= 57%

Two tailors X and Y are paid a total of Rs. 550 per week by their employer. If X is paid 120 percent of the sum paid to Y, how much is Y paid per week?

A. Rs. 200

B. Rs. 250

C. Rs. 300

D. None of these

Let the sum paid to Y per week Rs. Z

Then Z+120% of z=550

Z+(120/100) z=550

11/5 z=550

Z=(550X5)/11

Z=250

The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is:

A. 4.37%

B. 5%

C. 6%

D. 8.75%

Increase in 10 years=2,62,500-1,75,000=87500

Increase %= (87500/1,75,000)X100

=50%

Required avg. = (50/10) % =5%

Rajeev buys good worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pays sales tax @ 10%. Find the amount he will have to pay for the goods.

A. Rs. 6876.10

B. Rs. 6999.20

C. Rs. 6654

D. Rs. 7000

Rebate= 6 % of 6650= (6/100 X 6650)= Rs. 399

Sales tax= 10 % of Rs (6650-399)

=10/100X 6251= Rs. 625.10

Final amount=Rs. 6251+625.10= Rs 6876.10

In the new budget, the price of kerosene oil rose by 25%. By how much percent must a person reduce his consumption so that his expenditure on it does not increase?

Reduction in consumption= [(R/(100+R)) x 100]

=(25/125)x 100

= 20%

The population of a town is 1,76,400. If it increases at the rate of 5% per annum, what will be its population 2 years hence? What was it 2 years ago?

1. Population after *n* years = $P\left(1+\frac{R}{100}\right)^n$

2. Population *n* years ago = $\frac{P}{\left(1+\frac{R}{100}\right)^n}$

Population after 2 years= $176400 \times (1+5/100)^2$

$=176400 \times 21/20 \times 21/40$

$= 194481$

Population 2 years ago= $176400/(1+5/100)^2$

$=176400 \times 20/21 \times 20/21$

$= 160000$

Ex. 26. The value of a machine depreciates at the rate of 10% per annum. If its present value is Rs. 1,62,000, what will be its worth after 2 years? What was the value of the machine 2 years ago?

1. Value of the machine after *n* years = $P\left(1-\frac{R}{100}\right)^n$

2. Value of the machine *n* years ago = $\frac{P}{\left(1-\frac{R}{100}\right)^n}$

Value of the machine after 2 years= $[162000 \times (1-10/100)^2]$

$= 162000 \times (1-1/10)^2$

$= 162000 \times 9/10 \times 9/10$

$= \text{Rs. } 131220$

Value of the machine 2 years ago = $[162000 / (1- 10/100)^2]$

$= 162000 \times 10/9 \times 10/9$

$= 200000$

Ex. 27. During one year, the population of a town increased by 5% and during the next year, the population decreased by 5%. If the total population is 9975 at the end of the second year, then what was the population size in the beginning of the first year?

(Hotel Management, 2003)

Population in the beginning of the first year= $9975/[(1 + 5/100) (1 - 5/100)]$

$= 9975 \times (20/21) \times (20/19)$

$= 10000$

Ex. 29. If A's salary is 20% less than B's salary, by how much percent is B's salary more than A's ?

$[20/(100-20) \times 100]\% = 25\%$

Ex. 32. *In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% in both. Find the percentage of those who passed in both the subjects.*

Let A and B be the sets of students who failed in Hindi and English resp.

$n(A)=35$, $n(B)=45$, $n(A\cap B)=20$

$n(A\cup B)= n(A)+ n(B) - n(A\cap B)$

$= 35+ 45-20$

$= 60$

Hence percentage passed=100-60= 40%

Ex. 33. *In an examination, 80% of the students passed in English, 85% in Mathematics and 75% in both English and Mathematics. If 40 students failed in both the subjects, find the total number of students.*

6. AVERAGE

IMPORTANT FACTS AND FORMULAE

1. Average $\left(\dfrac{\text{Sum of observations}}{\text{Number of observations}}\right)$

2. Suppose a man covers a certain distance at x kmph and an equal distanc y kmph. Then, the average speed during the whole journey is $\left(\dfrac{2xy}{x+y}\right)$ kmp

Ex. 1. ***Find the average of all prime numbers between 30 and 50.***

Sol.. There are five prime numbers between 30 and 50.

They are 31, 37, 41, 43 and 47.

$\therefore$ Required average $= \left(\dfrac{31+37+41+43+47}{5}\right) = \dfrac{199}{5} = 39.8.$

Ex. 2. ***Find the average of first 40 natural numbers.***

Sol. Sum of first n natural numbers $= \dfrac{n(n+1)}{2}$

So, sum of first 40 natural numbers $= \dfrac{40 \times 41}{2} = 820.$

$\therefore$ Required average $= \dfrac{820}{40} = 20.5.$

Ex. 3. ***Find the average of first 20 multiples of 7.***

Required Average= 7(1+2+3+….+20) / 20

= 7(20x21) / 20x2= 147/2 = 73.5

Ex. 4. *The average of four consecutive even numbers is 27. Find the largest of th numbers.*

Let the numbers be x , x+2, x+4, x+6

[X+(x+2) +(x+4) + (x+6)] / 4 = 27

4x+12/4 = 27

X+3=27

X=24

Largest number= x+6 = 24+6= 30

Ex. 5. *There are two sections A and B of a class, consisting of 36 and 44 students respectively. If the average weight of section A is 40 kg and that of section B is 35 kg, find the average weight of the whole class.*

Total weight of (36+44) students= (36x40 + 44X35)kg=2980kg

Average weight of the whole class= 2980/80= 37.25kg

Ex. 6. *Nine persons went to a hotel for taking their meals. Eight of them spent Rs. 12 each on their meals and the ninth spent Rs. 8 more than the average expenditure of all the nine. What was the total money spent by them ?*

Let the average expenditure of all the nine be Rs. X

12x8+ (x+8) = 9x

8x=104

X=13

Total money spent= 9x= 9x13= Rs. 117

Ex. 7. *Of the three numbers, second is twice the first and is also thrice the third. If the average of the three numbers is 44, find the largest number.*

Let the third number be x, then second number=3x, first number=3x/2

X+3x+3x/2= 44x3

11x/2= 44x3

X= 24

Largest number=second number=3x=72

Ex. 8. *The average of 25 results is 18. The average of first twelve of them is 14 and that of last twelve is 17. Find the thirteenth result.*

Thirteenth result=(sum of 25 results)- (sum of 24 results)

= (25x18) – [(14x12)+(17x12)]

= 450-(168+204)

=450-372

= 78

Ex. 9. *The average of 11 results is 60. If the average of first six results is 58 and that of the last six is 63, find the sixth result.*

Sixth result= (58x6+63x6 - 60x11)

=66

Ex. 10. *The average weight of A, B, C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, find the weight of B.*

A+B+c= (45x3)kg=135kg

A+B= 40x2=80kg

B+C=43x2=86kg

B= (A+B)+(B+C)- (A+B+C)

= (80+86-135)kg

= 31 kg

Ex. 11. *The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher.*

Total age of 39 persons= 39x15= 585 years

Average age of 40 persons= 15 years 3 months= 15.25 years = 61/4 years

Total age of 40 persons= 61/4 x 40 = 610 years

Age of teacher = 610-585= 25 years

Ex. 12. *The average weight of 10 oarsmen in a boat is increased by 1.8 kg when one of the crew, who weighs 53 kg is replaced by a new man. Find the weight of the new man.*

Total weight increased=1.8x10=18 kg

Weight of new man =53+18= 71 kg

Ex. 13. *There were 35 students in a hostel. Due to the admission of 7 new students, the expenses of the mess were increased by Rs. 42 per day while the average expenditure per head diminished by Re 1. What was the original expenditure of the mess ?*

Let the original expenditure be Rs. X

Then

42(x-1)-35x=42

7x=84

X=12

Original expenditure= Rs. (35* 12)= Rs. 420

Ex. 14. *A batsman makes a score of 87 runs in the 17th inning and thus increases his average by 3. Find his average after 17th inning.*

Sol. Let the average after 17th inning = x

Then, average after 16th inning=x-3

16(x-3)+87=17x

X=39

Ex. 15. *Distance between two stations A and B is 778 km. A train covers the journey from A to B at 84 km per hour and returns back to A with a uniform speed of 56 km per hour. Find the average speed of the train during the whole journey.*

Average speed =(2xy)/(x+y)

= 67.2km/hr

21. SIMPLE INTEREST

IMPORTANT FACTS AND FORMULAE

1. **Principal** : The money borrowed or lent out for a certain period is called the ***principal*** or the ***sum***.
2. **Interest** : Extra money paid for using other's money is called ***interest***.
3. **Simple Interest (S.I.)** : If the interest on a sum borrowed for a certain period is reckoned uniformly, then it is called ***simple interest***.

Let Principal = P, Rate = R% per annum (p.a.) and Time = T years. Then,

(*i*) $\text{S.I.} = \left(\frac{P \times R \times T}{100}\right)$.

(*ii*) $P = \left(\frac{100 \times \text{S.I.}}{R \times T}\right)$; $R = \left(\frac{100 \times \text{S.I.}}{P \times T}\right)$ and $T = \left(\frac{100 \times \text{S.I.}}{P \times R}\right)$.

Ex. 1. ***Find the simple interest on Rs. 68,000 at $16\frac{2}{3}\%$ per annum for 9 months.***

Sol. P = Rs. 68000, $R = \frac{50}{3}\%$ p.a and $T = \frac{9}{12}$ years $= \frac{3}{4}$ years.

$$\therefore \quad S.I. = \left(\frac{P \times R \times T}{100}\right) = \text{Rs.}\left(68000 \times \frac{50}{3} \times \frac{3}{4} \times \frac{1}{100}\right) = \text{Rs. } 8500.$$

Ex. 2. ***Find the simple interest on Rs. 3000 at $6\frac{1}{4}\%$ per annum for the period from 4th Feb., 2005 to 18th April, 2005.***

Sol. Time = (24 + 31 + 18) days = 73 days $= \frac{73}{365}$ year $= \frac{1}{5}$ year.

P = Rs. 3000 and $R = 6\frac{1}{4}\%$ p.a. $= \frac{25}{4}\%$ p.a.

$$\therefore \quad S.I. = \text{Rs.}\left(3000 \times \frac{25}{4} \times \frac{1}{5} \times \frac{1}{100}\right) = \text{Rs. } 37.50.$$

Remark : The day on which money is deposited is not counted while the day on which money is withdrawn is counted.

Rupees 3600 is divided into two parts such that interest of one part at 5% per annum for 3 years in equal to the interest of second part at 6(1/4) % per annum for 4 years. Find the amount of the parts?

(Y x 5 x 3)/100 = [(3600-Y)x25x 4]/100x4

60Y= (3600-Y)x 100

60y=36000-100y

Y=36000/160

Y= 2250 1st part

3600-2250= 1350 2nd part

The amount received on a sum of rupees 2400 at certain rate for 4 years is rupees 3264. On increasing the rate from 1%, find the amount of the same principal for the same time?

SI=3264-2400=864

Rate= SI x 100 / PxT

=864x100 / 2400 x 4

= 864/96

= 9 % on increasing by 1%= 10%

SI =(2400x10x4)/100

= 960

A= 2400+960= 3360

3. A sum at simple interest at 13(1/2) % per annum amounts to Rs. 2502.50 after 4 years. Find the Sum.

Let the sum be Rs. x

SI= Rs. (x * 27/2 *4 *1/100)

=Rs. 27x/50

Amounts= x + 27x/50

=77x/50

77x/50=2502.50

X=2502.50*50 /77

X=1625

4. A Sum of Rs. 800 amounts to Rs. 920 in 3 years at simple interest. If the interest rate is increased by 3%, It would amount to how much?

SI= 920-800=Rs. 120, p=800, T=3

R=120x100 / 800x3 %

R=5%

New rate=5+3=8%

New SI= 800x8x3 / 100 = Rs. 192

New amount=800+192=Rs. 992

$\therefore$ New amount = Rs. (600 + 152) = Rs. 752.

Ex. 5. *Adam borrowed some money at the rate of 6% p.a. for the first two years, at the rate of 9% p.a. for the next three years, and at the rate of 14% p.a. for the period beyond five years. If he pays a total interest of Rs. 11, 400 at the end of nine years, how much money did he borrow?* (Bank P.O. 1999)

Sol. Let the sum borrowed be x. Then,

(X*6*2/100)+(x*9*3/100)+(x*14*4/100)=11400

3x/25 + 27x/100 + 14x/25 =11400

95x/100=11400

X=11400*100/95

X= Rs. 12000

Ex. 9. *The simple interest on a certain sum of money for $2\frac{1}{2}$ years at 12% per annum is Rs. 40 less than the simple interest on the same sum for $3\frac{1}{2}$ years at 10% per annum. Find the sum.*

$(x \times 10 \times 7)$ $(x \times 12 \times 5)$

7. AT What rate percent per annum will a sum of money double in 16 years?

8. The simple interest on a sum of money is 4/9 of the principal. Find the rate percent and time, if both are numerically equal.

Simple Interest Formula

The Formula for simple interest helps you find the interest amount if the principal amount, rate of interest and time periods are given.

Simple interest formula is given as:

$$SI = (P \times R \times T) / 100$$

Where SI = simple interest

P = principal

R = interest rate (in percentage)

T = time duration (in years)

In order t0 calculate the total amount, the following formula is used:

Amount (A) = Principal (P) + Interest (I)

Where,

Amount (A) is the total money paid back at the end of the time period for which it was borrowed.

Rishav takes a loan of Rs 10000 from a bank for a period of 1 year. The rate of interest is 10% per annum. Find the interest and the amount he has to the pay at the end of a year.

SI= 1000x10x1 / 100= 1000

A=10000+1000=11000

Namita borrowed Rs 50,000 for 3 years at the rate of 3.5% per annum. Find the interest accumulated at the end of 3 years.

SI= 50000x3.5x3 /100= Rs. 5250

Mohit pays Rs 9000 as an amount on the sum of Rs 7000 that he had borrowed for 2 years. Find the rate of interest.

Solution:

SI= 9000-7000=2000

R=SI x 100 / P x T

=2000x100 / 7000x2

= 14.29%

Mr Garg lent Rs 15000 to his friend. He charged 15% per annum on Rs 12500 and 18% on the rest. How much interest does he earn in 3 years?

ANSWER:

P=12500,T=3,R=15%
SI= 12500x15x3 /100 = 5625
Rest of the amount lent 15000-12500=2500
R=18%,T=3
SI= 2500x18x3 /100 =1350

Total SI=5625+1350= RS. 6975

Shikha deposited Rs 2000 in a bank which pays 6% simple interest. She withdrew Rs 700 at the end of first year. What will be her balance after 3 years?

P=2000,T=1,R=6%

SI=Rs. 120

Amount=2000+120=Rs. 2120

Withdrew 2120-700=Rs.1420

P=1420, R=6%,T=2

SI=1420x6x2 / 100 = 170.4

Total Amount after 3 years=Rs.1420+Rs.170.40 = Rs. 1590.40

Hari borrowed Rs 12600 from a moneylender at 15% per annum simple interest. After 3 years, he paid Rs 7070 and gave a goat to clear off the debt. What is the cost of the goat?

P=12600, R=15%, T=3

SI= Rs. 5670

A=P+SI= 12600+5670=Rs. 18270

18270-7070=Rs.11200

Divide Rs 3600 into two parts such that if one part be lent at 9% per annum and the other at 10% per annum, the total annum income is Rs 333.

ANSWER:

Let the first part be Rs. x. Second part =(3600−x)∴S.I. on x at 9% per annum for 1 years=x×9×1100=9x100And, S.I. on (3600−x) at 10% per annum =(3600−x)×1×101100=3600−x 10∴9x100+3600−x10=333=>9x+36000−10x100=333=>−x+36000=33300=>−x=33300−360 00=>−x=−2700=>x=2700First part = Rs. 2700 Second part =(3600−2700)=Rs. 900

If Rs 640 amounts to Rs 768 in 2 years 6 months, what will Rs 850 amount to in 3 years at the same rate per cent per annum?

ANSWER:

P=Rs. 640, A=Rs. 768, T=2 years 6 months =52 years S.I.=A−P =768−640 =Rs. 128 R=S.I.×100P×T=128×100×2640×5=8% P=Rs. 850, R=8%, T=3 years∴S.I. =P×R×T100=850×8×3100=204010=Rs. 204∴A=P+S.I. =8 50+204 =Rs. 1054

Compound Interest : Sometimes it so happens that the borrower and the lender agree to fix up a certain unit of time, say *yearly* or *half-yearly* or *quarterly* to settle the previous account.

In such cases, the amount after first unit of time becomes the principal for the second unit, the amount after second unit becomes the principal for the third unit and so on.

After a specified period, *the difference between the amount and the money borrowed is called the* ***Compound Interest*** *(abbreviated as C.I.) for that period.*

IMPORTANT FACTS AND FORMULAE

Let Principal = P, Rate = R% per annum, Time = n years.

I. When interest is compound Annually :

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^n$$

II. When interest is compounded Half-yearly :

$$\text{Amount} = P\left[1+\frac{(R/2)}{100}\right]^{2n}$$

III. When interest is compounded Quarterly :

$$\text{Amount} = P\left[1+\frac{(R/4)}{100}\right]^{4n}$$

IV. When interest is compounded Annually but time is in fraction, say $3\frac{2}{5}$ years.

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^3 \times \left(1+\frac{\frac{2}{5}R}{100}\right)$$

V. When Rates are different for different years, say $R_1\%$, $R_2\%$, $R_3\%$ for 1st, 2nd and 3rd year respectively.

$$\text{Then, Amount} = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right).$$

VI. Present worth of Rs. x due n years hence is given by :

$$\text{Present Worth} = \frac{x}{\left(1+\frac{R}{100}\right)^n}.$$

1. Find compound interest on Rs. 7,500 at 4% per annum for 2 years, compounded annually.

P= 7500, R=4%, T=2

Amount= Rs. $[7500x(1+4/100)^2]$

=7500 x 26/25 x 26/25

=Rs.8112

CI=8112-7500= Rs.612

2. Find Compound interest on Rs. 8,000 at 15% per annum for 2 years 4 months compounded annually.

Time = 2 years 4 months=2(4/12)years= 2(1/3) years

Amount=$8000x(1+15/100)^2$ x (1+ (1/3 x15) / 100)

=$8000x (1+3/20)^2$ x (1+1/20)

=8000x 23/20 x 23/20 x 21/20

= 11109

CI=11109-8000= Rs. 3109

3. Find Compound interest on Rs. 10,000 in 2 years at 4% per annum, the interest being compounded half yearly.

P=10,000, R=2 % per half year,

Time=2 years= 4 half years

Amount=10000 x $(1+2/100)^4$

= 10000 x $(51/50)^4$

= 10000x 51/50 x 51/50 x 51/50 x 51/50

= 10824.32

CI=10824.32-10000= Rs. 824.32

4. Find Compound interest on Rs.16,000 at 20% per annum for 9 months compounded quarterly.

P=16000 , T=9 months=3 quarters

R=20% per annum=5% per quarter

Amount=16000x$(1+5/100)^3$

=16000x 21/20 x 21/20 x 21/20

= 18522

CI=18522-16000=Rs.2522

5. If the simple interest on a sum of money at 5% per annum for 3 years is Rs. 1200, find the compound interest on the same sum for the same period at the same rate.

Rate=5% Time=3 SI=1200

P= 100x1200 /5x3 =8000

Amount=$8000x(1+5/100)^3$

$=8000x(21/20)^3$ =9261

CI=9261-8000= Rs. 1261

6. In what time will Rs. 1000 become Rs. 1331 at 10% per annum compounded annually?

P=1000, A=1331, R=10%

Let time be n years

$1000x(1+10/100)^n = 1331$

$(11/10)^n = 1331/1000$

n=3 years

7. If Rs. 500 amounts to Rs. 583.20 in two years compounded annually, find the rate of interest per annum.

P=500, A=583.20, Time = 2 years

Let the rate be R% per annum

$500(1+R/100)^2 = 583.20$

$(1+R/100)^2 = 5832/5000$

$(1+R/100)^2 = 11664/10000$

$(1+R/100)^2 = (108/100)^2$

$(1+R/100)= 108/100$

R= 8% p. a .

8. If the compound interest on a certain sum at 16 2/3 % for 3 years is Rs. 1270, find the simple interest on the same sum at the same rate and for the same period.

$CI =P(1+R/100)^n-P$

Let sum be x

$CI= x * (1+50/(3*100)^3 – x$

$=343x/216-x$

$=127x/216$

$127x/216=1270$

$X= 1270*216/127$

$X= 2160$

Sum is Rs.2160

SI= 2160x(50/3) x 3x 1/100 = Rs. 1080

9. The difference between the compound interest and simple interest on a certain sum at 10% for 2 years is Rs. 631. Find the sum.

Let sum be x

$CI = P(1+R/100)^n - P$

$= x(1+10/100)^2 - x = 21x/100$

$SI = x*10*2/100 = x/5$

$CI - SI = (21x/100) - (x/5) = 631$

$x/100 = 631$

$x = 63100$

10. Divide Rs. 1301 between A and B, so that the amount of A after 7 years is equal to the amount of B after 9 years, the interest being compounded at 4% per annum.

Let the two parts be Rs. X and (1301-x)

$X(1+4/100)^7 = (1301-x)(1+4/100)^9$

$x/(1301-x) = (1+4/100)^2$

$= 26/25 \times 26/25$

$625x = 676(1301-x)$

$X = 676$,

$1301 - 676 = 625$

11. A certain sum amounts to Rs. 7350 in 2 years and to Rs. 8575 in 3 years. Find the sum and rate percent.

12. A sum of money amounts to Rs. 6690 after 3 years and to Rs. 10,035 after 6 years on compound interest. Find the sum.